

High Dry Bean Intake and Reduced Risk of Advanced Colorectal Adenoma Recurrence among Participants in the Polyp Prevention Trial

Adequate fruit and vegetable intake was suggested to protect against colorectal cancer and colorectal adenomas; however, several recent prospective studies reported no association. We examined the association between fruits and vegetables and adenomatous polyp recurrence in the Polyp Prevention Trial (PPT). The PPT was a low-fat, high-fiber, high-fruit, and vegetable dietary intervention trial of adenoma recurrence, in which there were no differences in the rate of adenoma recurrence in participants in the intervention and control arms of the trial. In this analysis of the entire PPT trial-based cohort, multiple logistic regression analysis was used to estimate the odds ratio (OR) of advanced and nonadvanced adenoma recurrence within quartiles of baseline and change (baseline minus the mean over 3 y) in fruit and vegetable intake, after adjustment for age, total energy intake, use of nonsteroidal anti-inflammatory drugs, BMI, and gender. There were no significant associations between nonadvanced adenoma recurrence and overall change in fruit and vegetable consumption; however, those in the highest quartile of change in dry bean intake (greatest increase) compared with those in the lowest had a significantly reduced OR for advanced adenoma recurrence (OR = 0.35; 95% CI, 0.18–0.69; P for trend = 0.001). The median in the highest quartile of change in dry bean intake was 370% higher than the baseline intake. The PPT trial-based cohort provides evidence that dry beans may be inversely associated with advanced adenoma recurrence.